SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING
Academic Year: 2022-2023

Program: B Tech/MBA Tech Stream : (CE/CS) Year: I__ Semester: _II_____
Subject: Management Accounting for Engineers Time: 3 hrs. (11:00am to 2:00 pm)
Date: 17 /05 / 23 . No. of Pages: 04
Marks: 100 Marks

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) In all ___5_ questions to be attempted.
4) All questions carry equal marks.
5) Answer to each new question to be started on a fresh page.
6) Figures in brackets on the right hand side indicate full marks.
7) Assume Suitable data if necessary.

| Q-1 | | Answer briefly: | 20 |
|---|---|---|---|
| CO- 1; SO- ;<br>BL- Remember | a | Explain any five differences between Financial Accounting & Cost Accounting. | 5 |
| CO-4 ; SO- ;<br>BL- Evaluate | b | Given that, the total sales of a product is Rs. 1,00,000 and the fixed cost is Rs. 20,000 during the year 2022. If the product's variable cost and selling price per unit are Rs.15 and Rs.20 respectively, find out the P/V Ratio, BEP and Margin of Safety. | 5 |
| CO-2 ; SO- ;<br>BL- Evaluate | c | The following information is available in respect of material A:<br><br>Maximum usage - 300 units<br>Minimum usage – 200 units<br>Normal usage – 225 units<br>Time lag in procurement of material - Maximum 6 months , Minimum 2 months<br>Reorder quantity – 750 units<br><br>Calculate :<br>(a) Re-order level (c) Maximum level<br>(b) Minimum level (d) Average level | 5 |
| CO-2 ; SO- ;<br>BL- Understand | d | Explain the steps followed in Activity Based Costing. | 5 |

| | | | |
|---|---|---|---|
| Q-2<br>CO-2; SO-;<br>BL- Evaluate | a | Following are the details furnished by ABC Ltd. about its activity during the year ended 31st March, 2022. You are required to prepare a Cost Sheet showing various elements of cost and calculate the Profit.<br>Raw Material consumed - 40,000 units @ ₹ 7 per unit.<br>Direct Wages:<br>(a) Skilled worker ₹ 9 per unit.<br>(b) Unskilled worker ₹ 6 per unit.<br>Royalty on raw material consumed) @₹ 3 per unit.<br>Works overheads @₹ 8 per machine hour.<br>Machine Hours Worked 25,000.<br>Office Overheads @ 1/3rd of works cost<br>Sales Commission @₹ 4 per unit.<br>Units produced 40,000.<br>Stock of units at the end 4,000 units to be valued at cost of production per unit.<br>Selling price is ₹ 60 per unit. | 15 |
| Q-2<br>CO-2; SO-;<br>BL- Remember | b | Explain with examples, the classification of costs according to variability. | 5 |
| Q-3<br>CO-4; SO-;<br>BL- Evaluate | a | XYZ Ltd. manufactured and sold 10,000 units and 15,000 units in the First Year and the Second Year respectively. The selling price per unit was ₹ 60 in both the years. In the first year it suffered a loss of ₹ 30.000 and in the second year earned profit of ₹ 30,000. Considering the above information, calculate the followings:<br>a) Profit Volume Ratio<br>b) The amount of Fixed Cost<br>c) The BEP in value and units<br>d) Profit when 20.000 units are sold<br>e) The number of units to be sold to earn a profit of ₹ 1,00,000. . | 15 |
| Q-3<br>CO-4 SO-; BL-<br>Understand | b | What is Margin of Safety? Explain the calculation with an example. | 5 |

| | | | |
|---|---|---|---|
| Q-4<br>CO- 5; SO-;<br>BL-Apply | a | The following data is given:<br><br>**Product A** **Product B**<br>Direct materials Rs. 35 Rs. 20<br>Direct labour @ Rs. 2 per hour Rs. 8 Rs. 10<br>Variable overhead Rs. 9 Rs. 6<br>Selling price per unit Rs. 120 Rs. 110<br><br>State which product you would recommend to manufacture when:<br><br>a) Labour time is the key factor;<br>b) Sales value is the key factor;<br>c) Sales quantity is the key factor; | 15 |
| Q-4<br>CO- 4; SO-;<br>BL-Evaluate | b | The standard mix producing a product X is as follows:<br><br>Material A 60 tonnes @ Rs 10<br>Material B 90 tonnes @ Rs 20<br><br>**The actual mix was as follows:**<br><br>Material A 80 tonnes @ Rs 12<br>Material B 60 tonnes @ Rs 25<br><br>**Calculate all Material Variances.** | 5 |
| Q-5<br>CO-3; SO-;<br>BL-Apply | a | The details of expenses for production of 20,000 units at 100% capacity in a factory are given as follows:<br><br>**Particulars** ₹<br>Materials 5,00,000<br>Labour 4,00,000<br>Factory overhead (20% variable) 3,00,000<br>Office and administrative overhead (30% fixed) 2,50,000<br>Selling and distribution overheads (40% variable) 1,50,000<br><br>Prepare a Flexible Budget at 70% and 90% capacity level. | 15 |
| Q-5<br>CO-3; SO-;<br>BL- Evaluate | b | Define Budgeting and explain its benefits for a business organization. | 5 |

<table>
<tr>
<td>Q-6<br>CO- 5, SO-;<br>BL- Apply</td>
<td>a</td>
<td>The following are the particulars of two products P and Q:<br>
<table>
<tr><th></th><th>Product P<br>Rs. Per unit</th><th>Product Q<br>Rs. Per unit</th></tr>
<tr><td>Direct Materials</td><td>100</td><td>150</td></tr>
<tr><td>Direct Labour</td><td>140</td><td>240</td></tr>
<tr><td>Direct expenses</td><td>140</td><td>210</td></tr>
<tr><td>Selling price per unit</td><td>500</td><td>700</td></tr>
</table>
Total Fixed overheads is Rs.25000.<br>
The new sales manager has proposed the following two sales mixes<br>
1. Mix A – 300 units of P and 300 units of Q<br>
2. Mix B – 200 units of P and 400 units of Q.<br>
Advice the management as to which of the above TWO mixes are to be accepted.</td>
<td>12</td>
</tr>
<tr>
<td>Q-6<br>CO-1; SO-;<br>BL- Remember & Understand</td>
<td>b</td>
<td>Write Short Notes on:<br>
1. Accounting Concepts<br>
2. VED system of inventory classification</td>
<td>8</td>
</tr>
<tr>
<td>Q-7<br>CO-4; SO-;<br>BL- Evaluate</td>
<td>a</td>
<td>The following are the details for P Ltd for the month of Jan 2023.<br>
The Standard labour hours required for manufacture of one article of the finished product and the rate per hour are as under:<br>
<table>
<tr><th>Category of labour</th><th>Labour hours per unit</th><th>Rate per labour hour(Rs.)</th></tr>
<tr><td>Skilled</td><td>10</td><td>100</td></tr>
<tr><td>Semi-Skilled</td><td>7</td><td>50</td></tr>
<tr><td>Unskilled</td><td>12</td><td>20</td></tr>
</table>
The Actual production in a month was 100 articles.<br>
The details of actual hours worked by the different labour categories and the actual rate per labour hour are as under:<br>
<table>
<tr><th>Category of labour</th><th>Labour hours per unit</th><th>Rate per labour hour(Rs.)</th></tr>
<tr><td>Skilled</td><td>1020</td><td>110</td></tr>
<tr><td>Semi-Skilled</td><td>690</td><td>55</td></tr>
<tr><td>Unskilled</td><td>1220</td><td>19</td></tr>
</table>
Calculate the following Labour Variances from the given data:<br>
1. Labour cost variance.<br>
2. Labour rate variance<br>
3. Labour efficiency variance</td>
<td>12</td>
</tr>
<tr>
<td>Q-7<br>CO-2; SO-;<br>BL- Evaluate</td>
<td>b</td>
<td>Kuber Manufacturing Company estimates that its carrying cost is 15% and it's ordering cost is Rs. 9 per order. The estimated annual requirement is 48,000 units at a price of Rs. 4 per unit. Based on information provided respond to the following questions;<br>
i. What is the most economical number of units to order?<br>
ii. How many orders should be placed in a year?<br>
iii. How often should an order be placed?</td>
<td>8</td>
</tr>
</table>

**THE END**